
Introduction to Emerging Technology (ET)

— Chapter One —

Introduction

- **Technology:**
 - ✓ Comes from a Greek word *technologia* :
 - ❖ “Systematic treatment of an *art*, *craft*, or *techniques*”.
 - ✓ A combination of two Greek words “**Techno**” + “**logy**” :
 - ❖ “*Science* of the *mechanical* and *industrial arts*”.
- In general, **Technology** is the sum of any *techniques*, *skills*, *methods*, and *processes* used in the *production* of *goods* or *services* or in the *accomplishment* of *objectives* such as *scientific investigation*.

Definition of Emerging Technologies

- *Emerging technology* is a term generally used to describe a new technology.
- It may also refer to the continuing development of existing technology.
- *ET* commonly refers to **technologies that are currently developing**, or that are expected to be available within the next five to ten years.
- *Technological evolution* is a theory of **radical transformation** of **society** through **technological development**.
 - **Evolution** is the **process** of **developing** by **gradual changes**.

List of Recently Emerged Technologies

- *Artificial Intelligence*
- *Blockchain*
- *Augmented Reality and Virtual Reality*
- *Cloud Computing*
- *Internet of Things (IoT)*
- *Intelligent Apps (I-Apps)*
- *Big Data*
- *Robotic Process Automation (RPA)*

Industrial Revolution

- The **industrial revolution** was a period of *industrialization* and *innovation*, that took place during the late 1700s and early 1800s.
- Occurs when a society shifts from using tools to make products to use new source of energy.
 - ❖ Such as coal, to power machines in factories.
- It's started in England.

- The Most Important Inventions of the Industrial Revolution:

Transportation: The Steam Engine, The Railroad, The Diesel Engine, The Airplane.

Communication: The Telegraph. The Transatlantic Cable. The Phonograph. The Telephone.

Industry: The Cotton Gin. The Sewing Machine. Electric Lights.

Types of Industry

- The four types of industries are:
 1. **The primary industry**: involves *getting raw materials* e.g. mining, farming, and fishing.
 2. **The secondary industry**: involves *manufacturing* e.g. making cars and steel.
 3. **Tertiary industries**: provide a *service* e.g. teaching and nursing.
 1. **The quaternary industry**: involves *research* and *development* industries e.g. IT.

Industrial Revolution (IR 1.0) - 1760s

- Going from **hand production** methods to machines.
- The increasing use of **steam power**.
- The **development** of **machine tools**.
- The **rise of the factory system**.

Industrial Revolution (IR 2.0) - 1870s

- Also known as the **Technological Revolution**.
- Development of methods for **manufacturing interchangeable parts**.
- Widespread adoption of **pre-existing technological systems** such as **telegraph** and **railroad networks**.
- Introduction of ***Electric Power*** and ***Telephone***

Industrial Revolution (IR 3.0) - 1950s

- Also referred as *Digital Revolution*.
- The transition from mechanical and analog electronic technology to digital electronics.
- Mass production and widespread use of **digital logic circuits** and its derived technologies such as:
 - **Computer**,
 - **Handphones** and,
 - the **Internet**.

Fourth Industrial Revolution (IR 4.0)

- Emergent of **cyber-physical system**:
 - ✓ A mechanism that is controlled or monitored by computer-based algorithms, tightly integrated with the Internet and its users.
- Industries being widely use *Computer Numerical Control (CNC)* machines.
 - *CNC* are *operated* by *giving* it instructions using a computer.

- **IR 4.0** is the adoption of *Artificial Intelligence (AI)*.
 - ✓ Automated Robots, Internet of Things, additive manufacturing and autonomous vehicles.



Role of Data for Emerging Technologies (ET)

- **Data** is regarded as **the new oil** and **strategic asset** and it drives or even determines the future of:

Science, Technology, the economy ...

- **Understanding, exploring, and utilizing data** is reshaping the world and introducing a paradigm shift.
- **Data** is becoming the *world's most valuable asset, making innovation possible* and *transforming the way* we *make decisions* for the *future*.

Enabling device and networks for ET

- In the world of *digital electronic device* there are *four basic kinds of devices*:
 1. **Memory**: store information such as the contents of a spreadsheet or database.
 2. **Microprocessors**: execute software instructions to perform a wide variety of tasks such as running a word processing program or video game.

3. **Logic Devices:** provide specific functionalities, such as **device-to-device interfacing, data communication, signal processing, displaying data, timing and control operations**, and almost every other functionalities that a system must provide
4. **Networks:** Communication infrastructures and components for sending and receiving information

Human to Machine Interaction

- Refers to the *communication and interaction between a human and a machine* via a *user interface*.
- **HCI (Human-Computer Interaction)** is the study of how people interact with computers and to what extent computers are or are not developed for successful interaction with human beings.

Disciplines Contributing to HCI

- **Cognitive Psychology:** Limitations, information processing, performance prediction, cooperative working, and capabilities.
- **Computer Science:** Including graphics, technology, prototyping tools, UI management systems.
- **Linguistics:** Including natural language interfaces.
- **Engineering and Design:** Engineering principles and and graphic design.
- **Artificial Intelligence:** Intelligent softwares.
- **Human factors:** Display readability and Hardware design.

Future trends in Emerging Technology

5G Networks

Artificial Intelligence

Autonomous Devices

Blockchain

Augmented Analytics

Digital Twins

Immersive Experiences in Smart Spaces



End of Chapter One!